CURRICULUM AND SYLLABUS

FOR

B.E. DEGREE (COMPUTER SCIENCE & ENGINEERING) PROGRAMME

REGULATIONS 2020

CHOICE BASED CREDIT SYSTEM

FOR THE STUDENTS ADMITTED FROM THE

ACADEMIC YEAR 2020-2021 ONWARDS

Sri Ramakrishna Institute of Technology
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CURRICULUM STRUCTURE

SEMESTER – I

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1	20HSG01	Technical English – I	HS	2	0	2	4	3
2	20MHG01	Calculus and Linear Algebra	BS	3	1	0	4	4
3	20CHG01	Engineering Chemistry	BS	3	1	0	4	4
4	20ITG01	Programming for problem solving using C	ES	3	0	0	3	3
5	20EEG01	Basic Electrical and Electronics Engineering	ES	3	0	0	3	3
PRACTICALS								
6	20MEG01	Engineering Graphics	ES	0	0	4	4	2
7	20CHG02	Engineering Chemistry Laboratory	BS	0	0	3	3	1.5
8	20ITG02	Programming in C Laboratory	ES	0	0	4	4	2
			Total	**14**	**2**	**13**	**29**	**22.5**

SEMESTER – II

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1	20HSG02	Universal Human Values – II : Understanding Harmony	HS	3	0	0	3	3
2	20MHG02	Differential Equations and Complex Variables	BS	3	1	0	4	4

3	20PHG01	Engineering Physics	BS	3	1	0	4	4
4	20CSG01	Object Oriented Programming using C++	ES	3	0	0	3	3
5	20HMG02	Professional Ethics	HM	3	0	0	3	3
6	20AC001	Environmental Science and Engineering	AC	3	0	0	3	0
PRACTICALS								
7	20MEG02	Engineering Workshop	ES	0	0	4	4	2
8	20PHG02	Engineering Physics Laboratory	BS	0	0	3	3	1.5
9	20CSG02	Programming in C++ Laboratory	ES	0	0	4	4	2
			Total	**18**	**2**	**11**	**31**	**22.5**

SEMESTER – III

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1	20MHG06	Probability and Statistics (C)	BS	3	1	0	4	4
2	20ECG02	Digital Principles and System Design (C)	ES	3	0	0	3	3
3	20CS001	Computer Organization and Architecture (C)	PC	3	0	0	3	3
4	20CS002	Data Structure	PC	3	0	0	3	3
5	20IT002	Java Programming (C)	PC	3	0	0	3	3
6	20HSG03	Heritage of Tamils	HS	1	0	0	1	1
7		Open Elective I	OE	3	0	0	3	3
PRACTICALS								
8	20CS003	Data Structures Laboratory	PC	0	0	4	4	2
9	20IT004	Java Programming Laboratory (C)	PC	0	0	3	3	1.5
			Total	**18**	**1**	**7**	**26**	**23.5**

SEMESTER – IV

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1	20MHG04	Discrete Mathematics	BS	3	1	0	4	4
2	20IT007	Database Management System (C)	PC	3	0	0	3	3
3	20IT008	Operating Systems (C)	PC	3	0	0	3	3
4	20CS004	Software Engineering (C)	PC	3	0	0	3	3
5	20CS005	Analysis of Algorithms	PC	3	0	0	3	3
6		Open Elective II	OE	3	0	0	3	3
7	20AC002	Constitution of India	AC	3	0	0	3	0
8	20HSG04	Tamils and Technology	HS	1	0	0	1	1
PRACTICALS								
9	20IT009	Database Management System Laboratory (C)	PC	0	0	3	3	1.5
10	20IT010	Operating Systems Laboratory (C)	PC	0	0	3	3	1.5
			Total	**21**	**1**	**6**	**28**	**23**

SEMESTER – V

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1	20CS006	Theory of Computation	PC	3	1	0	4	4
2	20CS007	Object Oriented Analysis and Design (C)	PC	3	0	0	3	3
3	20CS008	Design of Computer Networks	PC	3	0	0	3	3
4	20CS009	Artificial Intelligence	PC	3	1	0	4	4
5		Professional Elective I	PE	3	0	0	3	3
6		Open Elective III	OE	3	0	0	3	3
PRACTICALS								
7	20CS010	Networks Laboratory	PC	0	0	4	4	2
8	20CS011	CASE Tools Laboratory (C)	PC	0	0	3	3	1.5
			Total	**18**	**2**	**7**	**27**	**23.5**

*(C) – Common with B.E.(CSE) & B.TECH (IT) Programme

SEMESTER – VI

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1	20CS012	Principles of Compiler Design	PC	3	1	0	4	4
2	20CS013	Full Stack Programming	PC	3	0	0	3	3
3		Professional Elective II	PE	3	0	0	3	3
4		Open Elective IV	OE	3	0	0	3	3
PRACTICALS								

5	20CS014	Compiler Design Lab	PC	0	0	4	4	2
6	20CS015	Full Stack Programming Laboratory	PC	0	0	4	4	2
7	20CS901	Design Project	EC	0	0	8	8	4
			Total	**12**	**1**	**16**	**29**	**21**

SEMESTER – VII

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1	20CS016	Cloud Computing Techniques	PC	3	0	0	3	3
2		Professional Elective III	PE	3	0	0	3	3
3		Professional Elective IV	PE	3	0	0	3	3
4		Professional Elective V	PE	3	0	0	3	3
5	20CS902	Final Year Project - I	EC	0	0	6	6	3
			Total	**12**	**0**	**6**	**18**	**15**

SEMESTER – VIII

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
THEORY								
1		Professional Elective VI	PE	3	0	0	3	3
2	20CS903	Final Year Project - II	EC	0	0	16	16	8
			Total	**3**	**0**	**16**	**19**	**11**

TOTAL NUMBER OF CREDITS: 160

PROFESSIONAL ELECTIVES

ELECTIVE – I [NETWORKING]

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1	20CSP11	Mobile Networks	PE	3	0	0	3	3
2	20CSP12	Storage Area Networks	PE	3	0	0	3	3
3	20CSP13	Wireless Sensor Networks	PE	3	0	0	3	3
4	20CSP14	Mobile Computing	PE	3	0	0	3	3
5	20CSP15	Software Defined Networks	PE	3	0	0	3	3
6	20CSP16	Social Network Analysis	PE	3	0	0	3	3
7	20CSP17	Distributed Systems	PE	3	0	0	3	3

ELECTIVE – II [INFORMATION SECURITY]

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1	20CSP21	Cryptography and Network Security	PE	3	0	0	3	3
2	20CSP22	Cyber Forensics	PE	3	0	0	3	3
3	20CSP23	Network and Web Security	PE	3	0	0	3	3
4	20CSP24	Ethical Coding and Risk Management	PE	3	0	0	3	3
5	20CSP25	Python Programming	PE	3	0	0	3	3
6	20CSP26	Problem Solving using Python Programming	PE	2	0	2	4	3

ELECTIVE – III [HUMAN COMPUTER INTERACTION]

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1	20CSP31	Augmented Reality and Virtual Reality	PE	3	0	0	3	3
2	20CSP32	Design Oriented HCI	PE	3	0	0	3	3
3	20CSP33	User Interface Design	PE	3	0	0	3	3
4	20CSP34	Human Computer Interaction	PE	3	0	0	3	3
5	20CSP35	Image Processing	PE	3	0	0	3	3
6	20CSP36	Internet of Things	PE	3	0	0	3	3

ELECTIVE – IV [SOFTWARE PROJECT ENGINEERING]

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1	20CSP41	Software Project Management	PE	3	0	0	3	3
2	20CSP42	Agile Software Development	PE	3	0	0	3	3
3	20CSP43	Software Requirements Engineering	PE	3	0	0	3	3
4	20CSP44	Software Quality Assurance and Testing	PE	3	0	0	3	3
5	20CSP45	Robotic Process Automation	PE	3	0	0	3	3

ELECTIVE – V [RECENT COMPUTING ADVANCEMENTS I]

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1	20CSP51	Salesforce	PE	3	0	0	3	3
2	20CSP52	Fundamentals of AWS	PE	3	0	0	3	3
3	20CSP53	Machine Learning & Deep Learning Techniques	PE	3	0	0	3	3
4	20CSP54	Entrepreneurship Development	PE	3	0	0	3	3
5	20CSP55	Soft Computing	PE	3	0	0	3	3
6	20CSP56	Data Mining	PE	3	0	0	3	3
7	20CSP57	Block chain Technology and Applications	PE	3	0	0	3	3

ELECTIVE – VI [RECENT COMPUTING ADVANCEMENTS II]

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1	20CSP61	Intellectual Property Rights	PE	3	0	0	3	3
2	20CSP62	Microsoft Azure	PE	3	0	0	3	3
3	20CSP63	Big Data Analytics	PE	3	0	0	3	3
4	20CSP64	Basics of Tensor Flow and Keras	PE	3	0	0	3	3
5	20CSP65	Machine Vision	PE	3	0	0	3	3
6	20CSP66	Foundation Skills in Integrated Product Development	PE	3	0	0	3	3
7	20CSP67	Information Security	PE	3	0	0	3	3

OPEN ELECTIVES

S. NO.	COURSE CODE	COURSE NAME	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
OFFERED BY B.E (ELECTRONICS AND COMMUNICATION ENGINEERING) PROGRAMME								
1	20ECE01	Electronic Measurements and Instrumentation	OE	3	0	0	3	3
2	20ECE02	Microcontrollers and its Applications	OE	3	0	0	3	3
3	20ECE03	Introduction to Embedded Systems	OE	3	0	0	3	3
4	20ECE04	Nano Electronics and Sensors	OE	3	0	0	3	3
5	20ECE05	Principles of VLSI Systems	OE	3	0	0	3	3
OFFERED BY B.E (ELECTRICAL AND ELECTRONICS ENGINEERING) PROGRAMME								
1	20EEE01	Energy Management Systems	OE	3	0	0	3	3
2	20EEE02	Medical Instrumentation	OE	3	0	0	3	3
3	20EEE03	PLC Programming	OE	3	0	0	3	3
4	20EEE04	Renewable Energy Systems	OE	3	0	0	3	3
5	20EEE05	Virtual Instrumentation & Data Acquisition	OE	3	0	0	3	3
6	20EEE06	Electric Vehicles	OE	3	0	0	3	3
OFFERED BY B.E (MECHANICAL ENGINEERING) PROGRAMME								
1	20MEE01	Automotive Fundamentals	OE	3	0	0	3	3
2	20MEE02	Computer Aided Design	OE	3	0	0	3	3
3	20MEE03	Introduction to Power Plant Engineering	OE	3	0	0	3	3

4	20MEE04	Introduction to Robotics	OE	3	0	0	3	3
5	20MEE05	3D Printing	OE	3	0	0	3	3